

Computer Organization Final Exam

هذا الملف تم حلّه من قبل الطالب: ليث عمرو.

هذه الحلول هي جهد بشري قابل للخطأ،
إن أصبت فمن الله، وإن أخطأت فمني ومن الشيطان.

هذا العمل مقدم ثوابه لروح والدي (قاسم مفيد عمرو)
فمن وجد فيه فائدة فليدع له بالرحمة والمغفرة.

Problem #1: Choose the correct answer. [7 * 1pt]

1. How many bits are represented by the constant X"56" in VHDL?

- a. 4 bits
- b. 16 bits
- c. 2 bits
- d. None of the above**

Sol = 8 bits

2. In VHDL, how are multiple process blocks executed?

- a. They run in a sequential way
- b. Only one process runs at a time
- c. They run one after another
- d. They run concurrently**

3. What is the correct order of sections in a VHDL file?

- a. Entity declaration, Library declaration, Architecture, Configuration
- b. Architecture, Entity declaration, Library declaration, Configuration
- c. Configuration, Library declaration, Architecture, Entity declaration
- d. Library declaration, Entity declaration, Architecture, Configuration**

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4. How is code executed inside a VHDL process?
 - a. Concurrently (parallel)
 - b. Sequentially (one after another)**
 - c. Randomly
 - d. Only during simulation

5. How is code executed outside a VHDL process?
 - a. Concurrently (parallel)**
 - b. Sequentially (one after another)
 - c. Only at compile time
 - d. Never executed

6. What VHDL element is used for communication between processes or entities?
 - a. Variable
 - b. Signal**
 - c. Constant
 - d. Port

7. Which VHDL element defines a reusable submodule for another design?
 - a. Entity
 - b. Package
 - c. Component**
 - d. Function

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Problem #2: The following VHDL code describes a digital circuit. Based on the code, draw the corresponding logic circuit. [10pts]

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity mux2to1 is
  Port ( a0, a1, s : in  STD_LOGIC;
        y : out STD_LOGIC );
end mux2to1;

architecture Structural of mux2to1 is

  component and_gate Port ( a : in  STD_LOGIC; b : in  STD_LOGIC; y : out STD_LOGIC );
    end component;

  component or_gate Port ( a : in  STD_LOGIC; b : in  STD_LOGIC; y : out STD_LOGIC );
    end component;

  component not_gate Port ( a : in  STD_LOGIC; y : out STD_LOGIC );
    end component;

  signal sn, d0, d1 : STD_LOGIC;

begin

  NOT1: not_gate port map ( a => s, y => sn );

  AND1: and_gate port map ( a => sn, b => a0, y => d0 );

  AND2: and_gate port map ( a => s, b => a1, y => d1 );

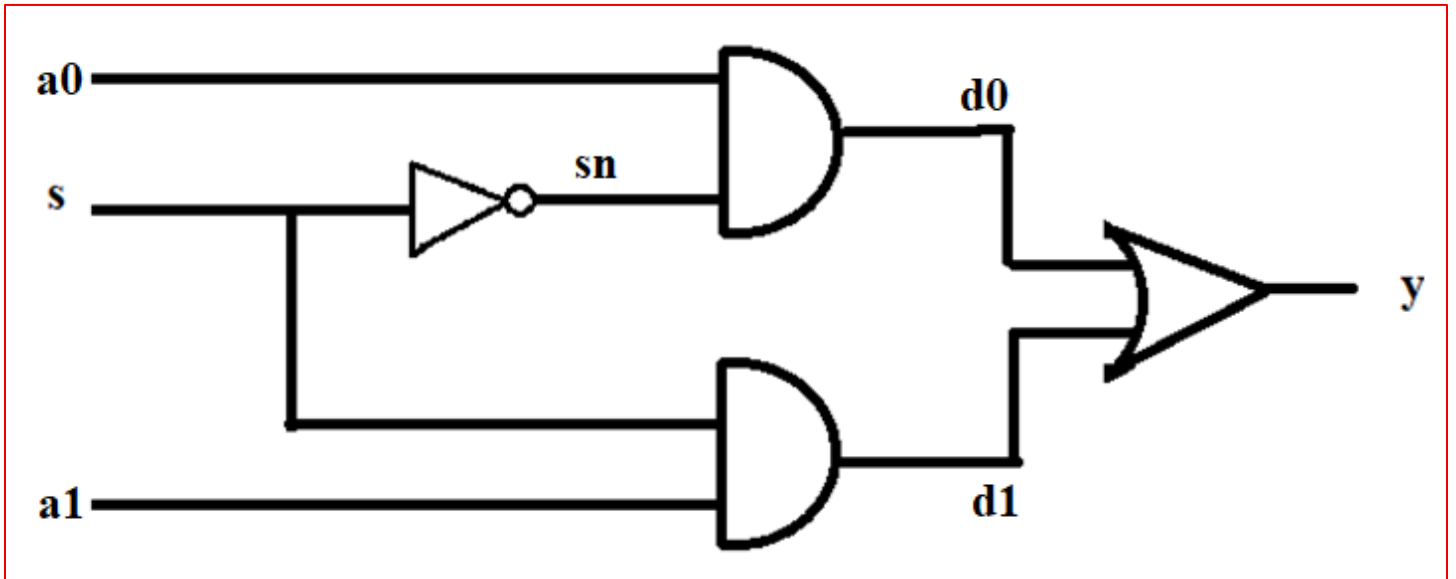
  OR1: or_gate port map ( a => d0, b => d1, y => y );

end Structural;
```

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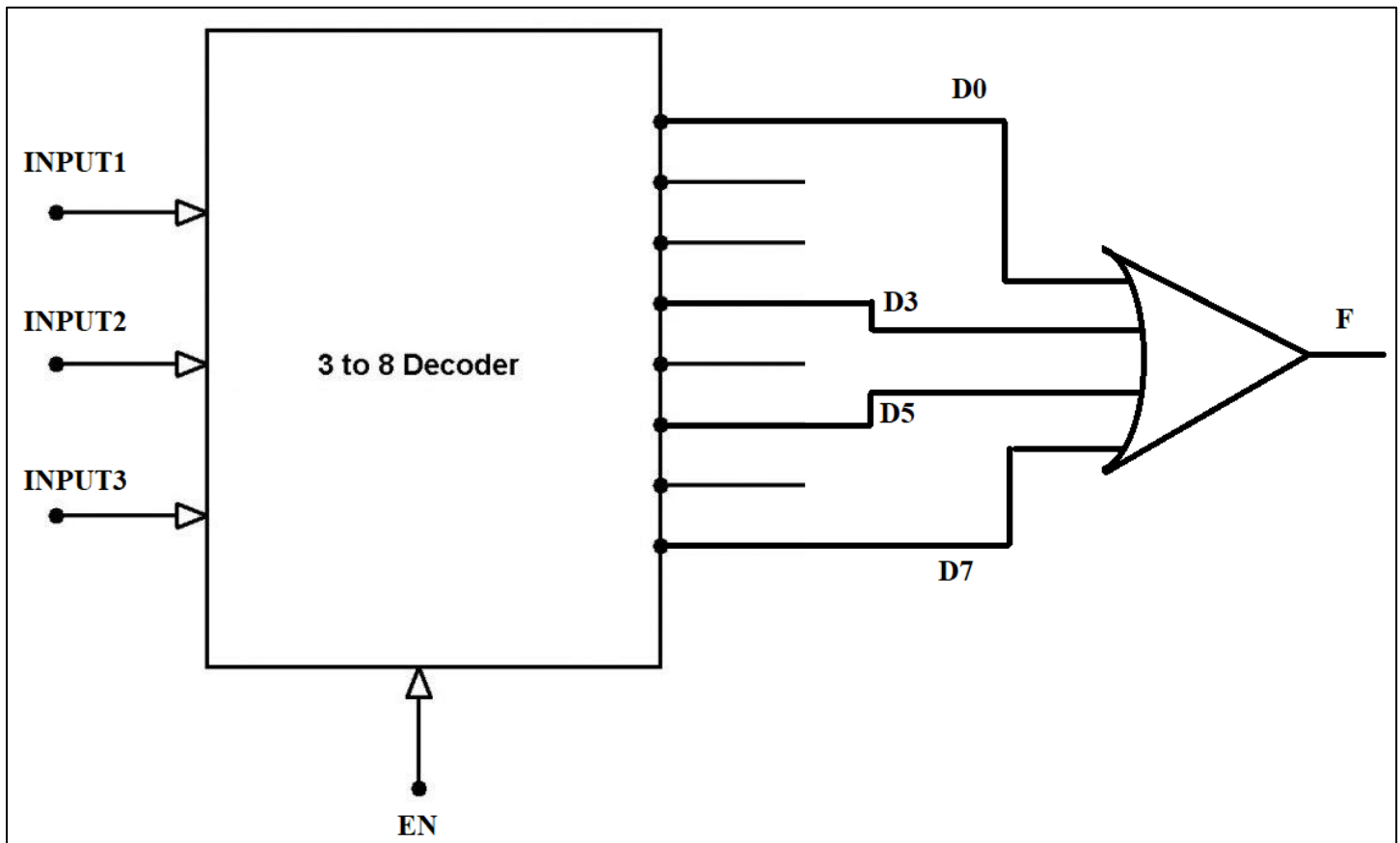
Solution:



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Problem #3: Design a 3-to-8 decoder in VHDL with an enable input. The decoder has INPUT has 3bits and 8 outputs (D0 to D7). [10pts]



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Solution (Behavioral):

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity decoder3x8_with_or is
    Port ( INPUT : in STD_LOGIC_VECTOR(2 downto 0);
          EN      : in STD_LOGIC;
          F       : out STD_LOGIC );
end decoder3x8_with_or;

architecture Behavioral of decoder3x8_with_or is
    signal D : STD_LOGIC_VECTOR(7 downto 0); -- Decoder outputs
begin

    process(INPUT, EN)
    begin
        D <= (others => '0'); -- Default: all outputs off
        if EN = '1' then
            case INPUT is
                when "000" => D(0) <= '1';
                when "001" => D(1) <= '1';
                when "010" => D(2) <= '1';
                when "011" => D(3) <= '1';
                when "100" => D(4) <= '1';
                when "101" => D(5) <= '1';
                when "110" => D(6) <= '1';
                when "111" => D(7) <= '1';
                when others => D <= (others => '0');
            end case;
        end if;
    end process;

    F <= D(0) or D(3) or D(5) or D(7);

end Behavioral;
```

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Solution (Dataflow):

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity decoder3x8_with_or is
    Port (
        INPUT : in STD_LOGIC_VECTOR(2 downto 0);
        EN    : in STD_LOGIC;
        F      : out STD_LOGIC
    );
end decoder3x8_with_or;

architecture Dataflow of decoder3x8_with_or is
    signal D : STD_LOGIC_VECTOR(7 downto 0);
begin

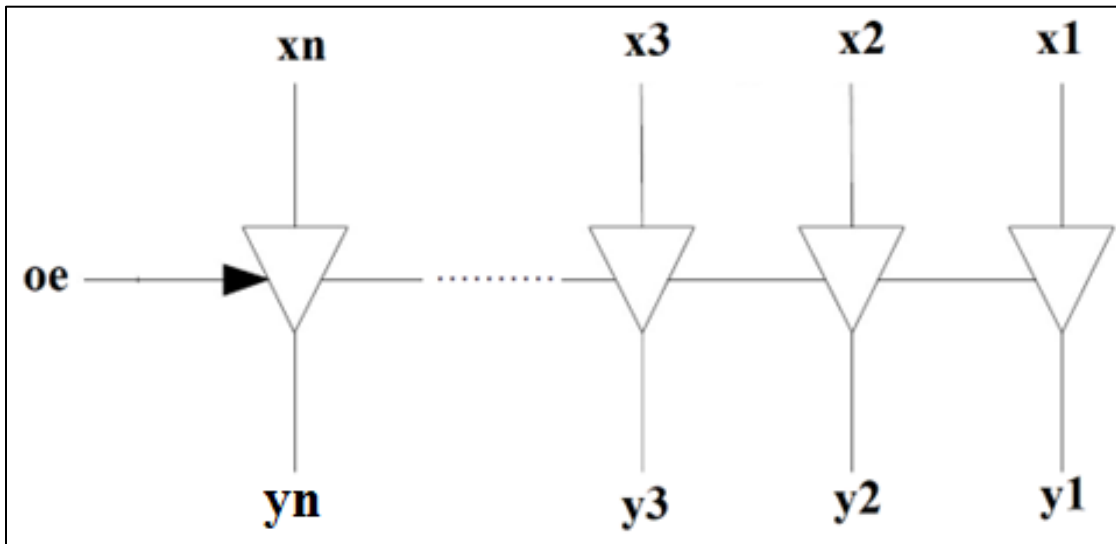
    D(0) <= '1' when (EN = '1' and INPUT = "000") else '0';
    D(1) <= '1' when (EN = '1' and INPUT = "001") else '0';
    D(2) <= '1' when (EN = '1' and INPUT = "010") else '0';
    D(3) <= '1' when (EN = '1' and INPUT = "011") else '0';
    D(4) <= '1' when (EN = '1' and INPUT = "100") else '0';
    D(5) <= '1' when (EN = '1' and INPUT = "101") else '0';
    D(6) <= '1' when (EN = '1' and INPUT = "110") else '0';
    D(7) <= '1' when (EN = '1' and INPUT = "111") else '0';

    F <= D(0) or D(3) or D(5) or D(7);

end Dataflow;
```

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Problem #4: Write the VHDL code for this N-bit buffer using a generic for N with a given appendix for **buffer1_bit.vhd** (last page) . Then write a testbench to test the buffer. [15 pts]



buffer_nbit.vhd

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity buffer_nbit is
    generic ( N : integer := 8 );
    port ( x : in  STD_LOGIC_VECTOR(N-1 downto 0);
          oe : in  STD_LOGIC;
          y : out STD_LOGIC_VECTOR(N-1 downto 0) );
end buffer_nbit;

architecture Structural of buffer_nbit is
    component buffer_1bit port ( x : in  std_logic; oe : in  std_logic; y : out std_logic );
    end component;
begin
    gen_buffer: for i in 0 to N-1 generate
        bit_buffer: buffer_1bit
            port map ( x => x(i), oe => oe, y => y(i) );
    end generate gen_buffer;
end Structural;
    
```

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tb_buffer_nbit.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity tb_buffer_nbit is
end tb_buffer_nbit;

architecture Behavioral of tb_buffer_nbit is
    constant N : integer := 4;
    component buffer_nbit
        generic ( N : integer );
        port ( x : in STD_LOGIC_VECTOR(N-1 downto 0);
              oe : in STD_LOGIC;
              y : out STD_LOGIC_VECTOR(N-1 downto 0) );
    end component;

    signal x_tb : STD_LOGIC_VECTOR(N-1 downto 0);
    signal oe_tb : STD_LOGIC;
    signal y_tb : STD_LOGIC_VECTOR(N-1 downto 0);
begin

    uut: buffer_nbit generic map (N=> N) port map ( x => x_tb, oe => oe_tb, y => y_tb );

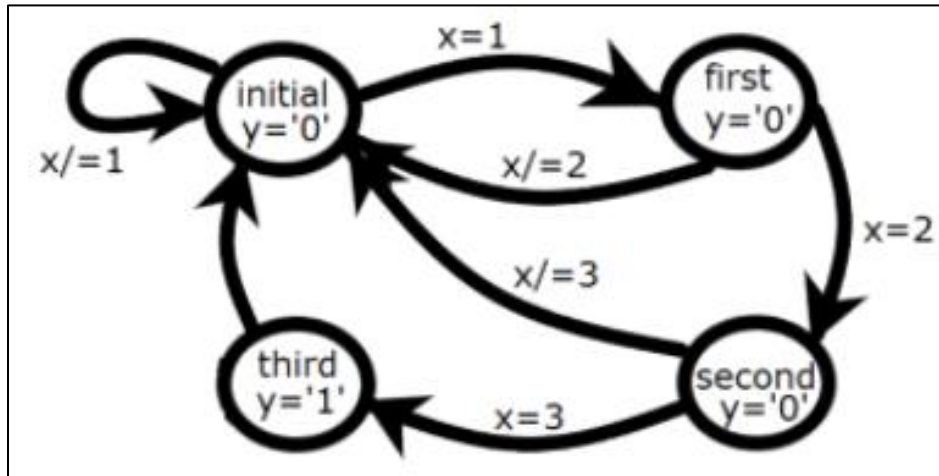
    stim_proc: process
    begin
        x_tb <= "1010";
        oe_tb <= '0';
        wait for 10 ns;

        -- Test case 2: Output enabled
        oe_tb <= '1';
        wait for 10 ns;
    end process;
end Behavioral;
```

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Problem #5: Write VHDL code to implement the following Finite State Machine (FSM) diagram, where the output y will be '1' if the value of three consequent inputs of input x is 1-2-3, which represents a password. [10 pts]



Solution (One Process)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity fsm_password is
    Port ( clk, rst : in STD_LOGIC;
          x  : in STD_LOGIC_VECTOR(1 downto 0);
          y  : out STD_LOGIC );
end fsm_password;

architecture Behavioral of fsm_password is
    type state_type is (S0, S1, S2, S3);
    signal state : state_type;
begin
```

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```
process(clk, rst)
begin
  if rst = '1' then
    state <= S0;
  elsif rising_edge(clk) then
    case state is
      when S0 =>
        if x = "01" then    -- x = 1
          state <= S1;
        else
          state <= S0;
        end if;
      when S1 =>
        if x = "10" then    -- x = 2
          state <= S2;
        else
          state <= S0;
        end if;
      when S2 =>
        if x = "11" then    -- x = 3
          state <= S3;
        else
          state <= S0;
        end if;
      when S3 =>
        state <= S0;
    end case;
  end if;
end process;

y <= '1' when state = S3 else '0';

end Behavioral;
```

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Solution (Two Process)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity fsm_password is
  Port (
    clk, rst : in  STD_LOGIC;
    x      : in  STD_LOGIC_VECTOR(1 downto 0);
    y      : out STD_LOGIC
  );
end fsm_password;

architecture Behavioral of fsm_password is
  type state_type is (S0, S1, S2, S3);
  signal current_state, next_state : state_type;
begin

  -- Process 1: State register (sequential)
  process(clk, rst)
  begin
    if rst = '1' then
      current_state <= S0;
    elsif rising_edge(clk) then
      current_state <= next_state;
    end if;
  end process;
```

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```
-- Process 2: Next state and output logic (combinational)
process(current_state, x)
begin
    -- Default assignments
    next_state <= current_state;
    y <= '0';

    case current_state is
        when S0 =>
            if x = "01" then    -- x = 1
                next_state <= S1;
            end if;

        when S1 =>
            if x = "10" then    -- x = 2
                next_state <= S2;
            else
                next_state <= S0; -- Reset on wrong input
            end if;

        when S2 =>
            if x = "11" then    -- x = 3
                next_state <= S3;
            else
                next_state <= S0; -- Reset on wrong input
            end if;

        when S3 =>
            y <= '1';          -- Output pulse
            next_state <= S0;   -- Auto-reset after success
    end case;
end process;

end Behavioral;
```

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Appendix: buffer_1bit.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity buffer_1bit is
    Port)
    x : in  STD_LOGIC; -- Input data
    oe : in  STD_LOGIC; -- Output enable
    y : out STD_LOGIC  -- Output
;
end buffer_1bit;

architecture Behavioral of buffer_1bit is
begin
    y <= x when oe = '1' else 'Z'; -- Tri-state logic
end Behavioral;
```

تم بفضل الله، اللهم ارحم عبدك (قاسم مفيد عمرو) واجعله من ورثة جنّة النعيم.

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